

Bass Parameter Guide

Parameter reference — Thee Coast Machine

1. Internal Structure

Saw / Square oscillator + Sub oscillator -> Moog-style 4-pole Ladder Filter -> Drive -> LFO. TB-303 inspired, with sub-bass, chords, slide, accent and 4-voice polyphony.

Element	Role	Sonic character
Main oscillator	Saw (rich harmonics) or Square (hollow)	Fat vs. hard
Sub oscillator	Sine one octave below	Weight in the bottom
Ladder filter (Moog-style)	Lowpass 4-pole, capable of self-oscillation	Quack / squelch
Filter envelope	Sweeps the cutoff -- the "wow" of acid	TB-303 signature
Accent	Filter + amp boost on high-velocity notes	Accented notes pop
Glide	Portamento between notes (slide)	TB-303 "slide" feel
LFO	Cyclic modulation of cutoff or pitch	Wobble / vibrato
Bass Chord	Adds automatic harmony voices on top of one note	Maj / Min / Power / etc.

2. Parameter Reference

WAVE (Saw / Square)

Main waveform selection.

- **Saw:** rich harmonics, TB-303 signature
- **Square:** hollow / harder edge

Tip: Acid uses Saw. Square fits House pumping bass or 80s Synthwave.

CUTOFF (100 - 5000 Hz)

Ladder filter **cutoff**.

- **100-300 Hz:** muffled, dub
- **300-800 Hz:** standard round bass
- **800-2000 Hz:** TB-303 sweet spot
- **2000-5000 Hz:** filter nearly wide open

Tip: Interacts with EnvMod.

RESO (0 - 10)

Ladder filter **resonance**.

- **0-2**: no resonance
- **2-5**: natural push
- **5-8**: clear "quack"
- **8-10**: edge of self-oscillation

Tip: For extreme acid, push past 9.5.

ENV MOD (0 - 6000 Hz)

Filter envelope **modulation depth**.

- **0**: static filter
- **500-2000**: subtle "wow"
- **2000-4000**: standard House
- **4000-6000**: full sweep, acid

Tip: TB-303: CUTOFF 500-800, RESO 8-10, ENV MOD 4000-5500.

DECAY (0.05 - 1.5 sec)

Bass envelope decay.

- **0.05-0.15**: acid pluck
- **0.15-0.30**: standard acid
- **0.30-0.60**: sustained House
- **0.60-1.5**: dub bass, sub pads

Tip: Longer DECAY = slower filter motion.

ACCENT (0 - 1)

Accent boost on high-velocity notes.

- **0.0**: accent disabled
- **0.3-0.5**: subtle emphasis
- **0.5-0.8**: clear accent
- **0.8-1.0**: dramatic (Hardfloor)

GLIDE (0 - 0.2 sec)

Portamento between notes.

- **0.0-0.005**: no glide
- **0.01-0.04**: natural slide
- **0.04-0.10**: gooey
- **0.10-0.20**: dramatic SFX

Tip: Triggers only on notes marked "Slide".

DRIVE (0 - 1)

Saturation on output.

- **0.0:** clean
- **0.1-0.3:** subtle weight
- **0.3-0.6:** acid signature
- **0.6-1.0:** industrial bass

Tip: TB-303 sweet spot: 0.4-0.7.

SUB (0 - 1)

Sub oscillator mix.

- **0.0:** no sub
- **0.2-0.4:** subtle weight
- **0.4-0.7:** solid sub
- **0.7-1.0:** sub-dominated

Tip: Verify on full-range monitors / headphones.

LFO RATE / DEPTH / TGT (Rate 0.05-20 Hz, Depth 0-1)

LFO modulation.

- **RATE 0.05-1 Hz:** slow drift
- **RATE 1-5 Hz:** vibrato
- **RATE 5-20 Hz:** tremolo
- **TGT = Cutoff:** filter wobble
- **TGT = Pitch:** pitch vibrato (+/- 1 semitone)

Tip: Classic acid wobble: Rate 0.2-0.5, Depth 0.3-0.5, Cutoff.

CHORD (Off / +Oct / Power / Maj / Min / Sus4)

Automatic harmony.

- **Off:** mono (default)
- **+Oct:** adds an octave up
- **Power:** adds a perfect 5th
- **Maj:** major 3rd + 5th
- **Min:** minor 3rd + 5th
- **Sus4:** 4th + 5th

Tip: Chord ON = 3 voices per note. 4-voice polyphony budget.

3. Character Presets

Goal	WAVE	CUT	RESO	ENV	DEC	DRIVE	SUB
TB-303 acid (classic)	Saw	600	9.5	5500	0.28	0.55	0.0
Hardfloor (extreme acid)	Saw	480	9.5	5500	0.16	0.75	0.10
Deep House (fat, deep)	Saw	360	2.5	1200	0.55	0.10	0.50
Detroit Techno	Saw	500	3.5	2200	0.40	0.20	0.30
Dub bass (heavy, long)	Saw	300	2.0	1000	0.80	0.15	0.65
Sub bass (Trap, 808)	Saw	380	2.0	1500	0.70	0.20	0.70
House square	Sq	500	4.0	2500	0.40	0.30	0.30
Industrial (distorted)	Saw	550	6.0	3200	0.20	0.85	0.20
Synthwave (retro)	Saw	450	3.0	1500	0.50	0.20	0.40
Footwork (tight sub)	Saw	400	2.5	1400	0.60	0.18	0.60

4. Tuning Order

#	Do this	Why
Step 1	Pick WAVE (Saw or Square)	Saw = acid / lead; Square = funky / retro
Step 2	Shape with CUTOFF + RESO	Squelchy, or round?
Step 3	Set ENV MOD for motion	Static or dynamic filter?
Step 4	DECAY for length	Fit it to the space in the track
Step 5	DRIVE for forwardness	Clean or grit
Step 6	SUB for low-end weight	Survives small speakers
Step 7	LFO / CHORD for flavour	Wobble and harmony

5. Troubleshooting

Symptom	Cause & fix
Bass feels "not there"	CUTOFF too low, SUB too high. Lower SUB, raise CUTOFF.
Kick and bass mud together	Raise SC AMT to 0.4-0.5, or separate CUTOFF from kick PITCH.
No acid "squelch"	Try RESO 8.5+, ENV MOD 4000+, DECAY 0.20-0.30.
Slide does not work	GLIDE too short. Start 0.03-0.06.
Chord sounds thin	Chord ON uses more voices. 4-voice limit.
Higher DRIVE = quieter	Tanh normalises level. Compensate with Bass Vol.
LFO inaudible	Check that LFO DEPTH is not 0.

※ The bass-kick interaction is the most important in the mix. Always design the bass against the kick. Separate CUTOFF from the kick PITCH, and use SC AMT to duck the bass under the kick - these two tricks resolve most low-end issues.